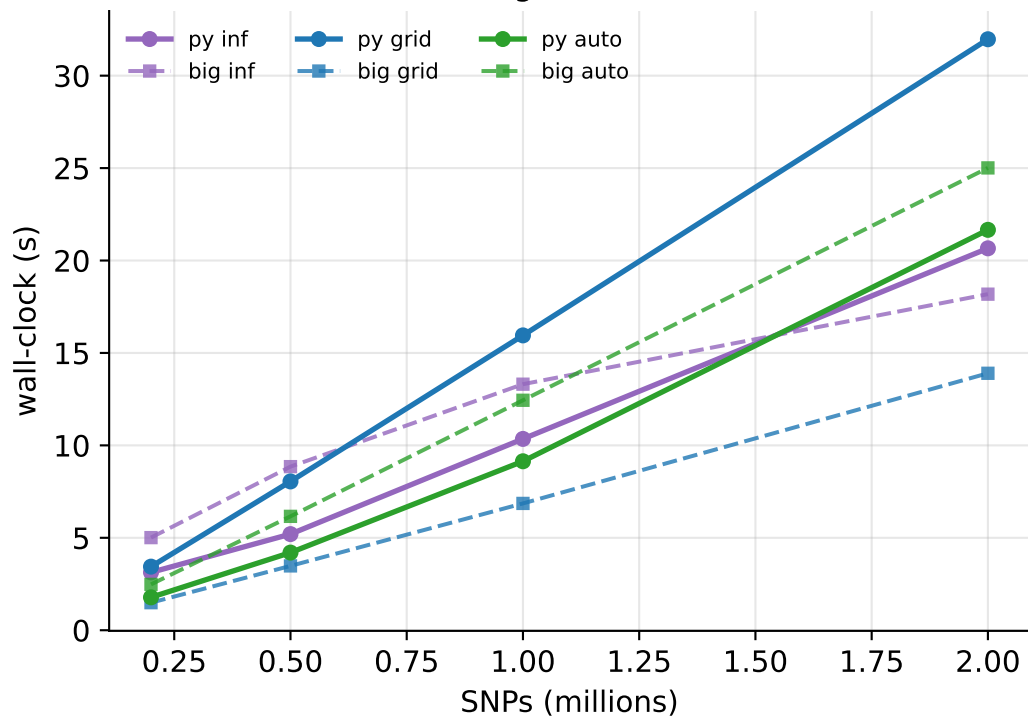
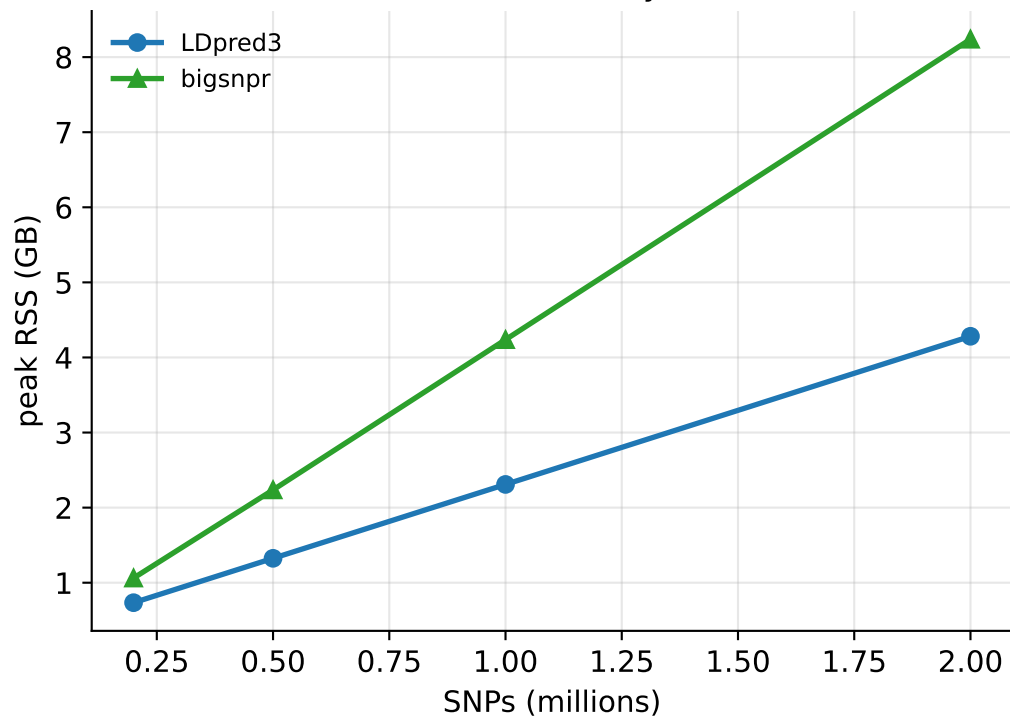


LDpred3 vs bigsnpr — realistic LD (coalescent), N=100k, $h^2=0.5$

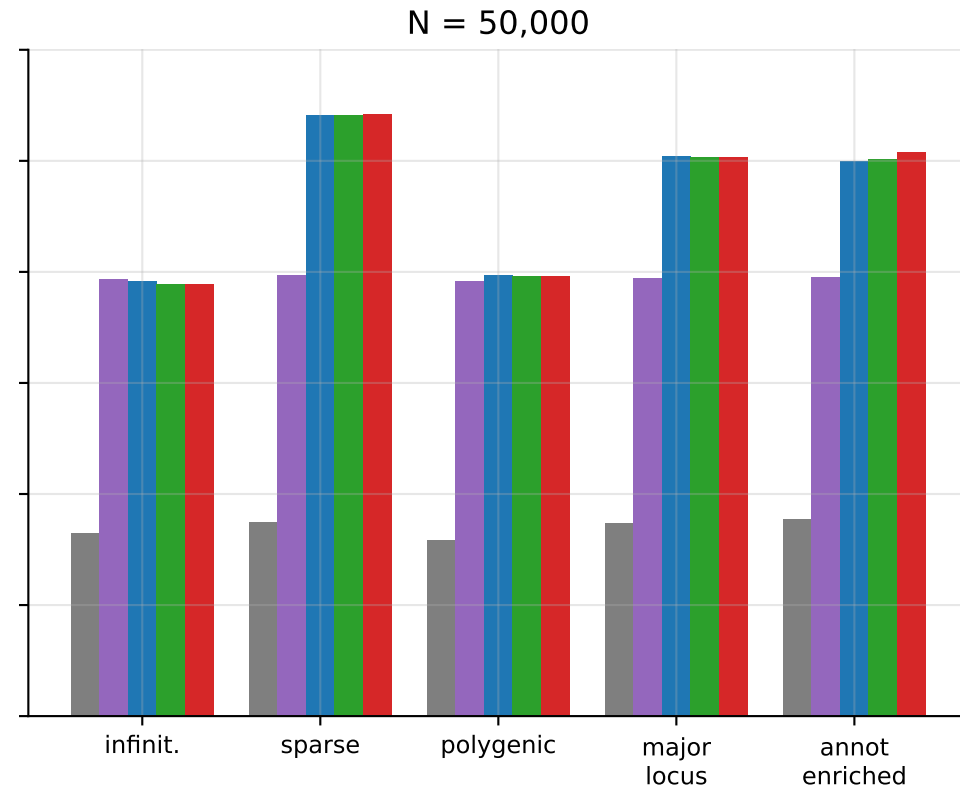
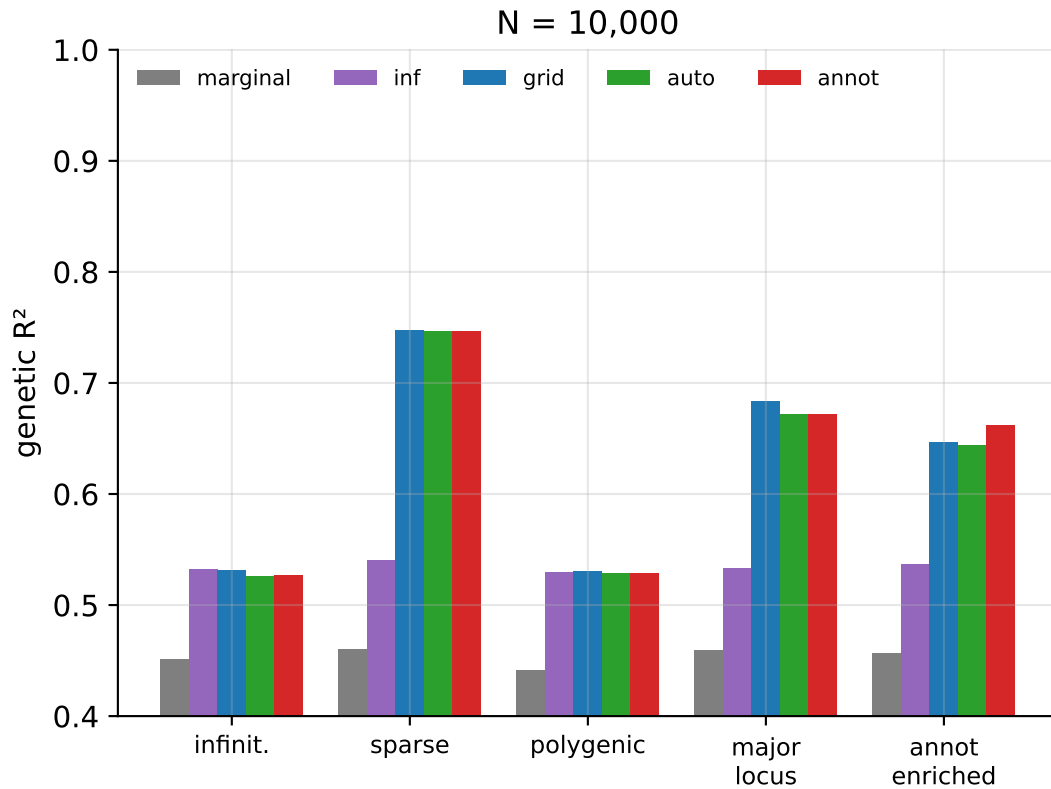
Running time, 1 core



Peak memory

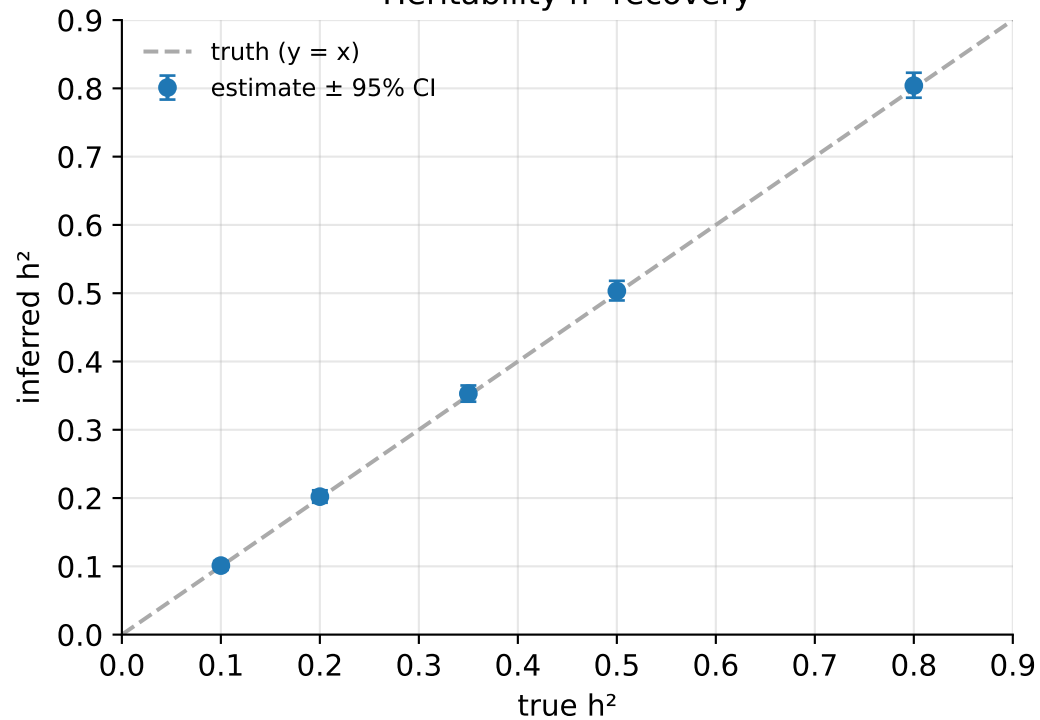


Accuracy by genetic architecture (realistic LD, $h^2=0.5$)

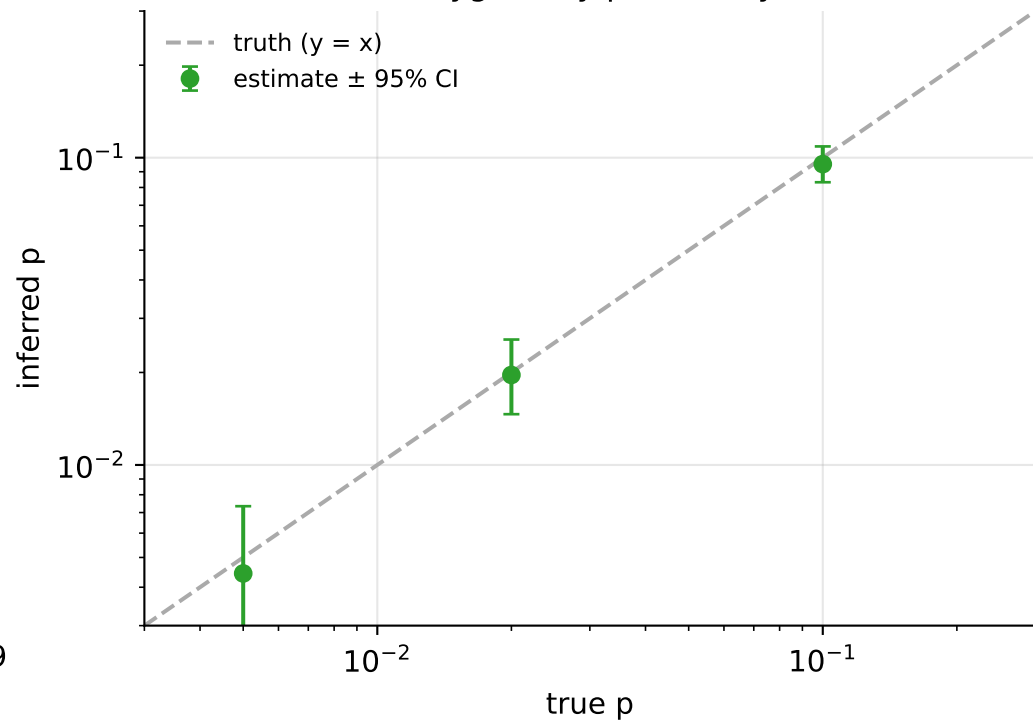


Inference evaluation (LDpred3-auto-infer, no validation cohort)

Heritability h^2 recovery

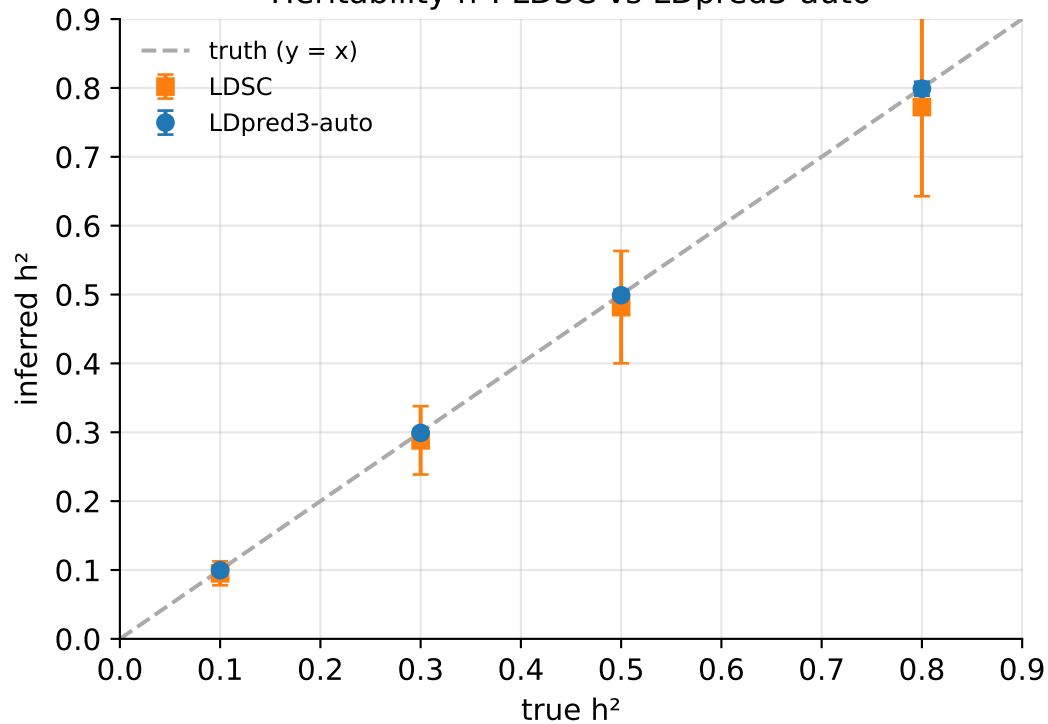


Polygenicity p recovery

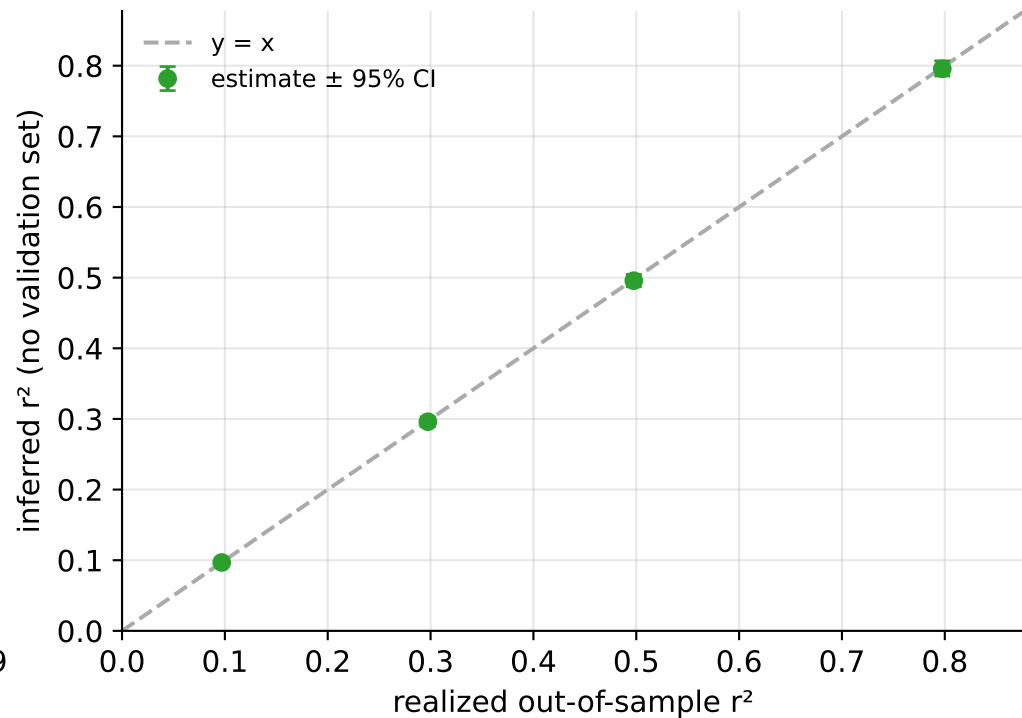


Inference cross-checks (realistic coalescent LD)

Heritability h^2 : LDSC vs LDpred3-auto

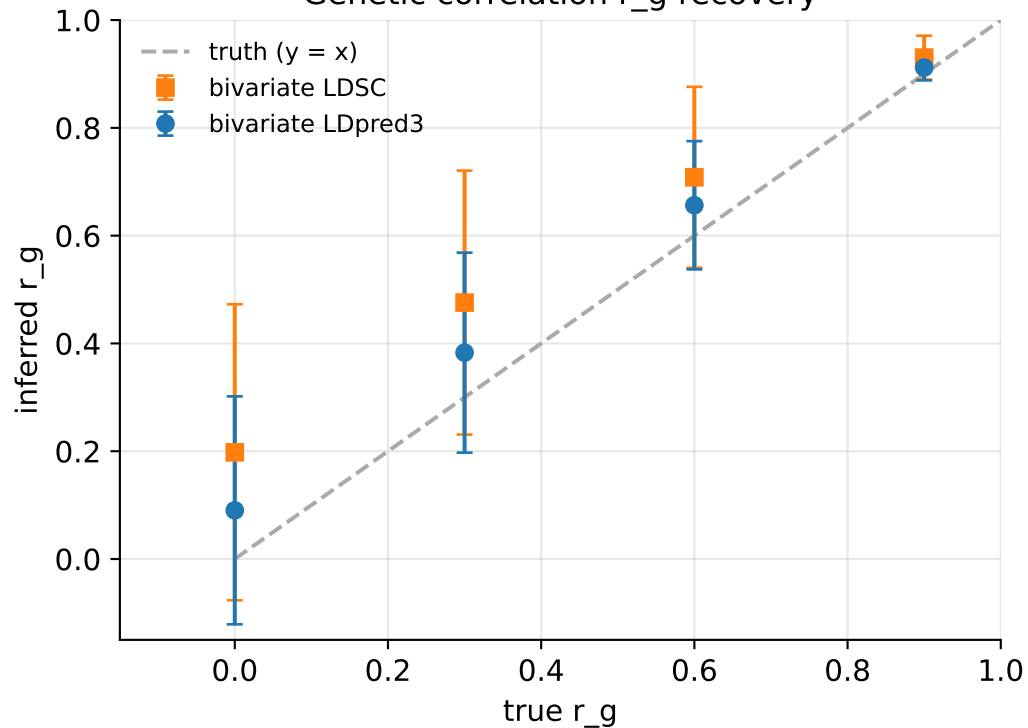


Predictive r^2 : estimated vs realized

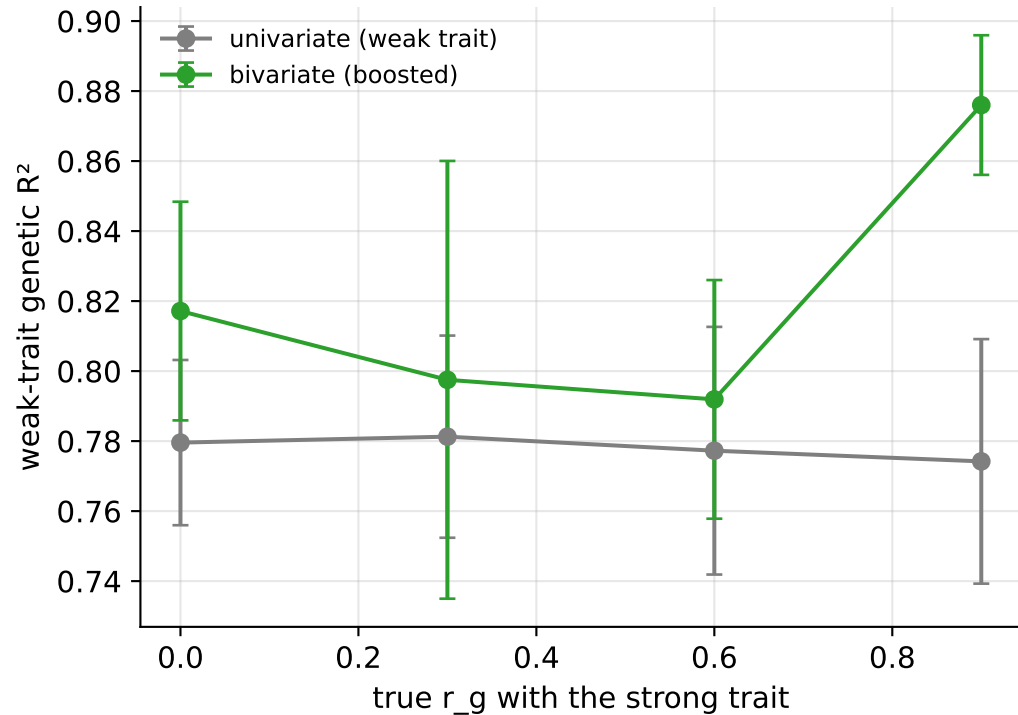


Bivariate analysis (realistic coalescent LD)

Genetic correlation r_g recovery

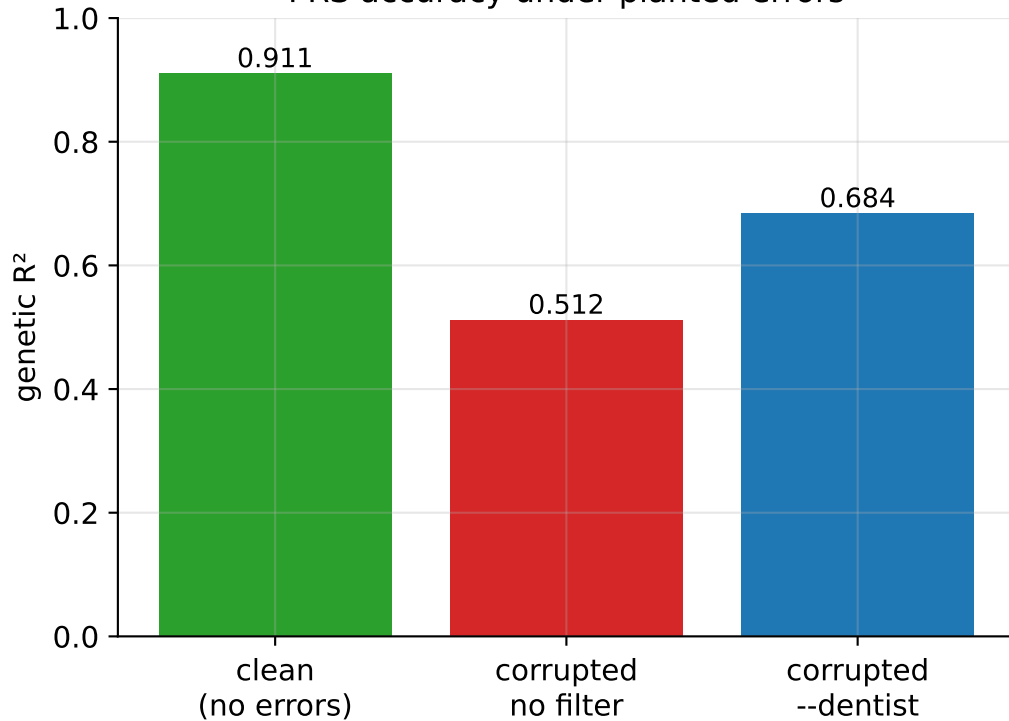


Weak-trait prediction gain ($N_2 \ll N_1$)

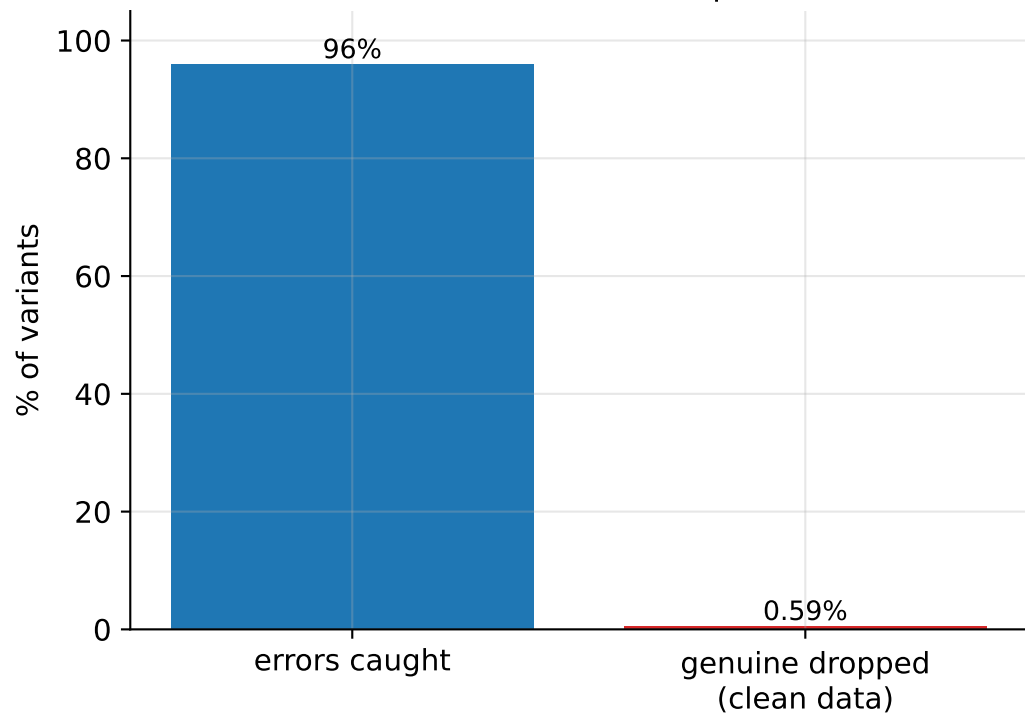


DENTIST LD-consistency filter (--dentist)

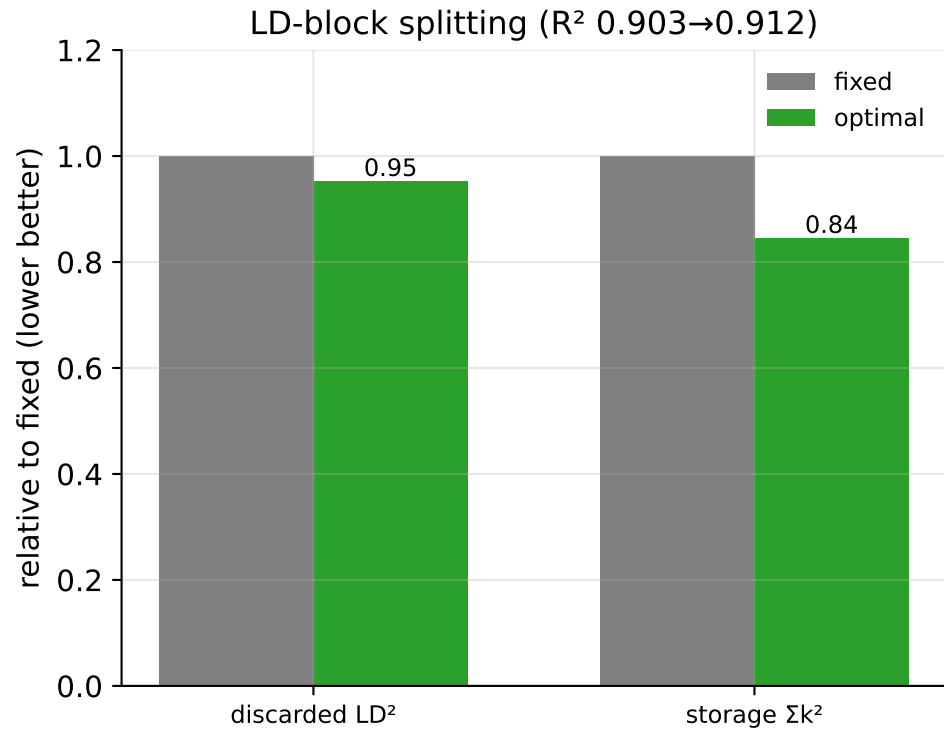
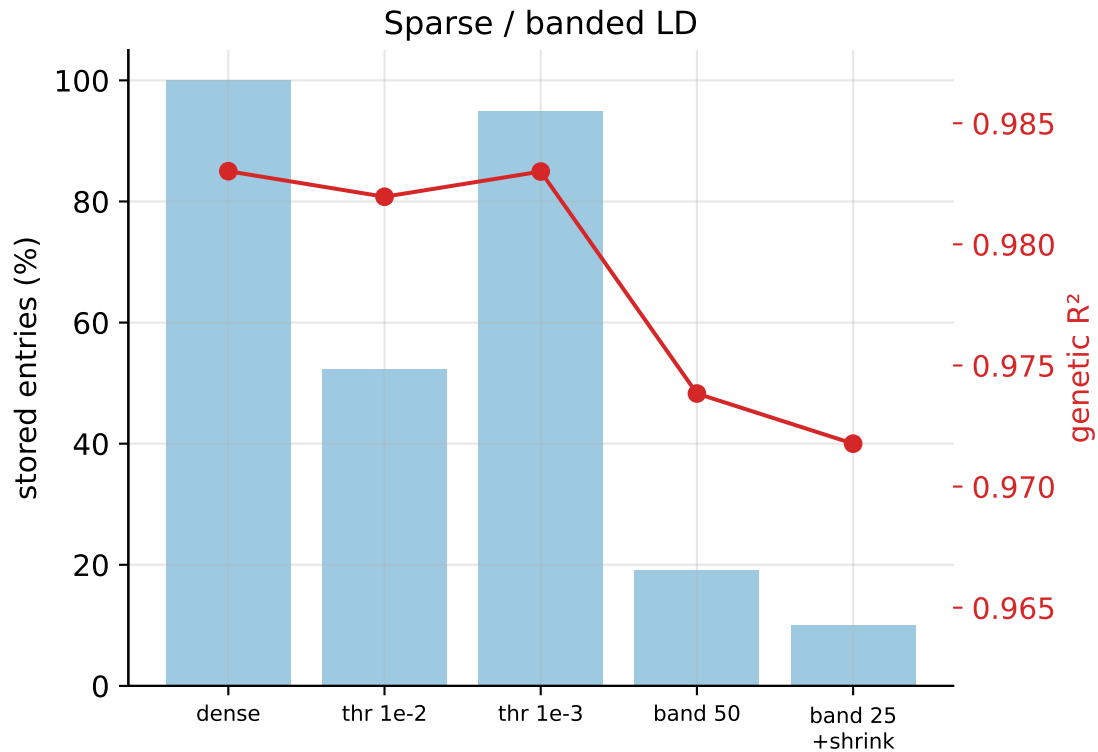
PRS accuracy under planted errors



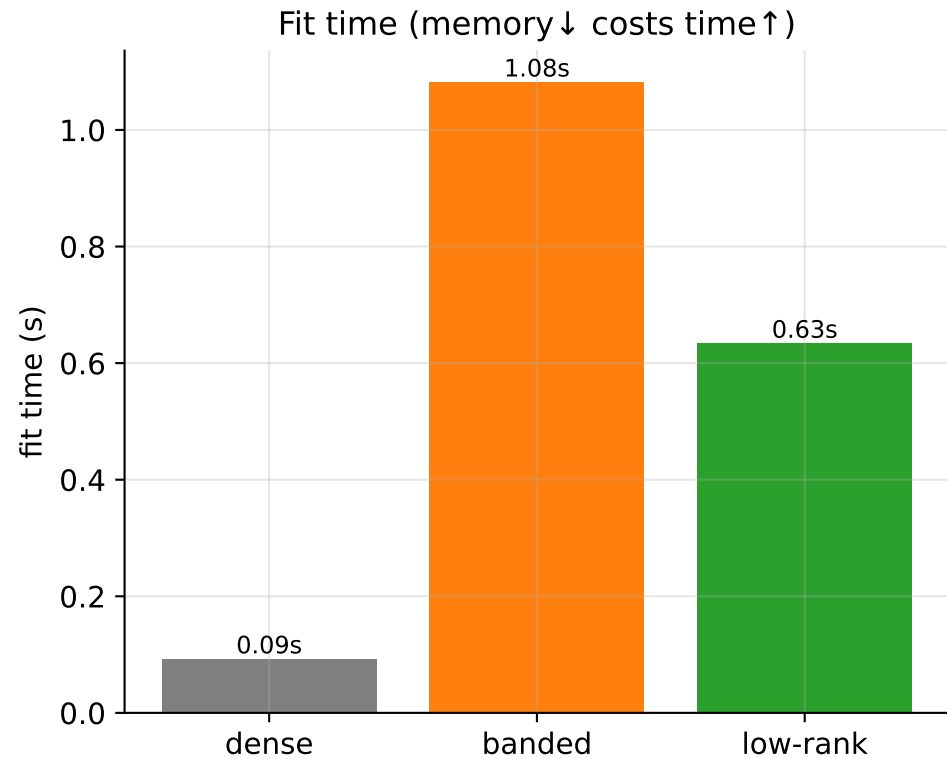
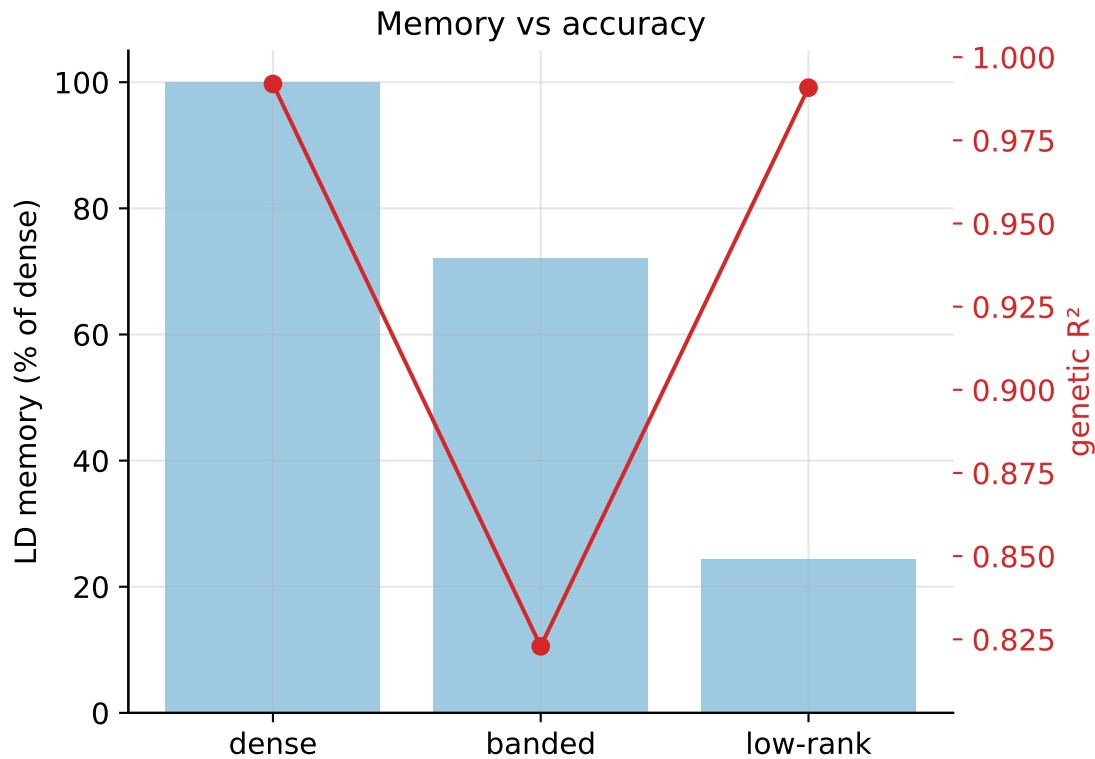
Catch rate vs false-drop cost



LD representation: storage vs accuracy



LD representations at scale — realistic coalescent LD (low-rank matches dense at ~1/4 memory; banding loses accuracy)



Performance

